



IndianOil

JOURNEY RISK MANAGEMENT (JRM) STUDY

Salem Terminal TO JAYAM & CO

Objective of the JRM Report

This JRM report is designed to ensure compliance with the Central Motor Vehicle Rules, 1989 (CMVR), AIS 140 standards, and the Road Transport Safety Policy (RTSP). It provides a comprehensive risk assessment for the transportation of hazardous materials along specified routes. By integrating these legal frameworks, the report offers a broad strategy for identifying and mitigating route-specific risks.

Regulatory Compliance

The report complies with the Central Motor Vehicles (Eleventh Amendment) Rules, 2022, mandating safe transportation practices for N2 and N3 category vehicles carrying hazardous materials. These rules require detailed route assessments, especially regarding road conditions, speed limits, and risk areas, to ensure safety compliance.

Risk Management Strategy

This report categorizes transportation routes into high-risk and medium-risk areas, with a focus on factors such as sharp turns, accident-prone regions, and elevation changes. The goal is to provide actionable

recommendations to minimize these risks, including speed regulations, driver warnings for hazardous zones, and the option of alternate routes.

Compliance with the Road Transport Safety Policy (RTSP)

The report integrates RTSP provisions, including mandatory driving hours, rest periods, and nighttime driving restrictions. It ensures that drivers follow official guidelines, such as taking prescribed rest breaks and avoiding dangerous road conditions like poor visibility, heavy crowds, or high-traffic areas during peak hours.

Emergency Preparedness and Response

The report highlights the significance of predetermined emergency stops for refueling, rest, and overnight stays. It includes protocols for safe responses to road hazards, alternative routes, and rerouting processes if roads are closed or severe weather arises. This aligns with the RTSP emphasis on driver safety and rapid emergency response.

Environmental Considerations

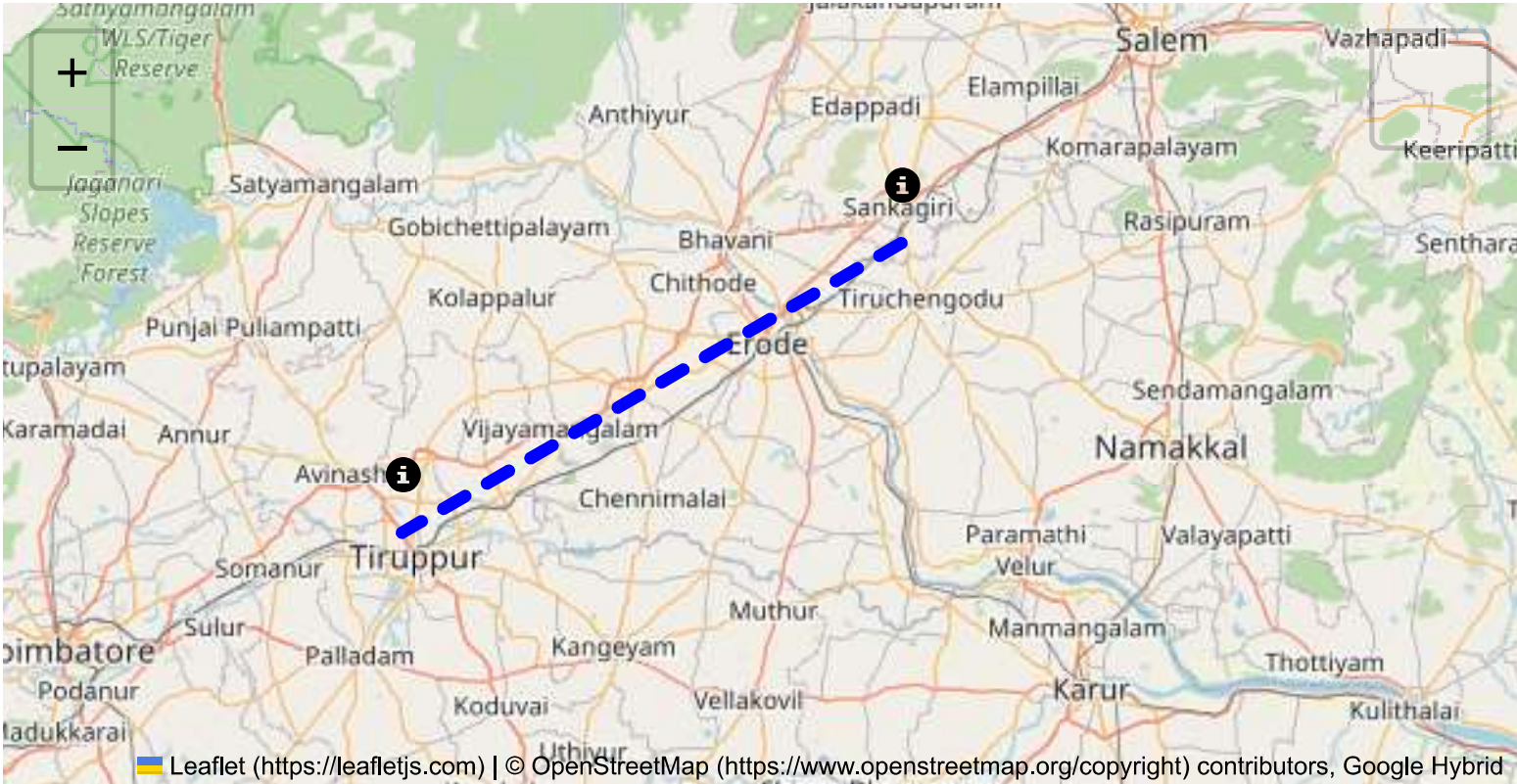
The JRM report addresses environmental risks along the route, ensuring compliance with environmental protection laws in ecologically sensitive zones. It suggests strategies such as identifying areas near water bodies, forests, or populated regions and implementing safety measures to minimize environmental impacts during transport.

Journey Risk Mitigation

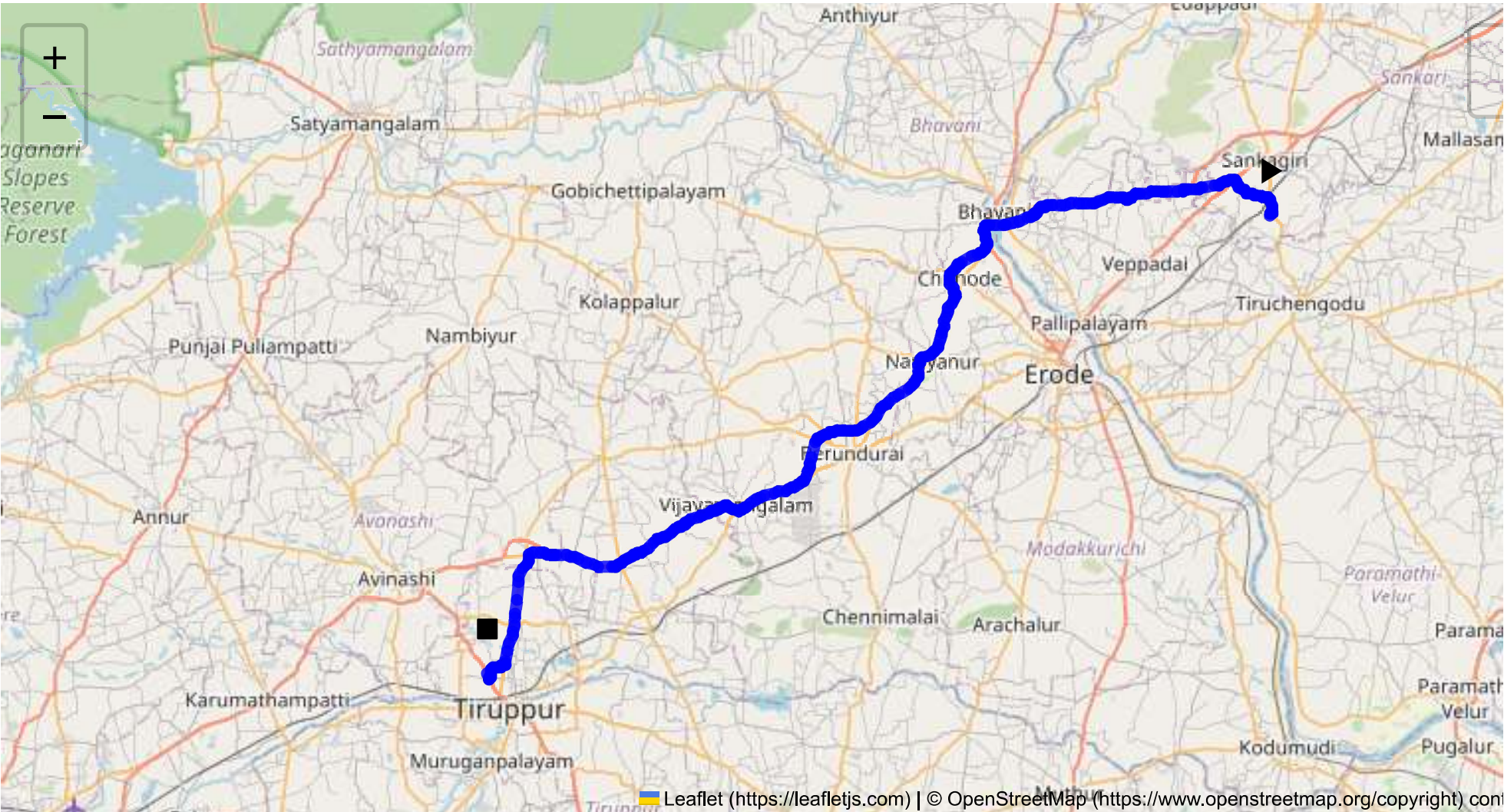
The report includes route-specific risk assessments, detailed journey charts, and defensive driving guidelines for each transport route. Integration with vehicle tracking systems guarantees real-time warnings on hazardous areas, speed limits, and mandatory stops, consistent with RTSP and CMVR safety norms.

Compliance with Government Directives

This report fully adheres to governmental directives regarding hazardous material transportation, implementing mandatory speed limits, nighttime driving restrictions, and comprehensive driver briefings and real-time alerts about route-related risks.



Route Summary:
Total Distance: 85.53 km
Estimated Duration: 1.8 hours
Adjusted Duration (Heavy Vehicle): 2.2 hours
Start: (11.4381, 77.8734)
End: (11.125996, 77.329278)



Welcome to the Journey Risk Management Study

Route Overview

The route from CVQF+23W, Sangagiri to SF 83, TS35/1, Avinashi - Tiruppur Rd in Tiruppur spans approximately 85.53 kilometers, primarily utilizing National and State Highways. The major roads involved include NH544 and local connecting roads. The journey typically takes around 1.77 hours for heavy vehicles carrying hazardous materials.

Weather Conditions

The region experiences a tropical climate with three primary seasons: summer (March to June), monsoon (July to October), and winter (November to February).

- **Summer:** High temperatures, occasionally reaching above 40°C. Heat might affect road conditions and vehicle performance.
- **Monsoon:** The area receives substantial rainfall, leading to potential flooding, particularly in low-lying areas.
- **Winter:** Mild weather, but fog can occur in the early morning hours, reducing visibility.

Traffic Patterns

- **Peak Hours:** Typically, peak traffic is observed in the morning (8:00 AM - 10:30 AM) and evening (5:00 PM - 8:00 PM).
- **Congestion-Prone Areas:** Areas such as Salem and Erode might experience severe traffic congestion during peak hours and market days.

Road Quality and Infrastructure

- **Road Conditions:** Major highways like NH544 are generally well-maintained. However, local connecting roads might have potholes and uneven surfaces.
- **Infrastructure:** Adequate signage but some areas may have limited lighting at night. Road widening and construction activities may cause temporary diversions.

Alternative Routes

For emergencies, alternative routes include:

- Via SH 86 cutting through smaller towns, though this may increase travel time.
- Bypass roads around major towns can serve as alternatives but require knowledge of local road conditions.

Local Regulations on Hazardous Material Transport

- **Regulations:** Adherence to the Central Motor Vehicles Rules for hazardous goods is mandatory, including proper labeling, vehicle safety measures, and emergency information systems.
- **Regulatory Stops:** Checkpoints may be in place to verify compliance with transport regulations.

Historical Incidents Involving Heavy Vehicles

- **Accidents:** Occasional incidents of vehicle overturns and spills primarily due to driver fatigue and sharp turns.
- **Incident Zones:** Areas near Salem and Erode have documented higher numbers of heavy vehicle-related incidents.

Environmental Considerations and Sensitive Areas

- **Areas of Concern:** Nearby wildlife sanctuaries and waterways require careful driving to minimize environmental impact.
- **Spill Response:** Swift action is critical near sensitive areas, demanding awareness of emergency cleanup protocols.

Communication Coverage

- **Coverage:** Generally good along major highways.
- **Dead Zones:** Possible signal drop in more rural stretches and ghat sections. Use satellite communication devices if frequent communication is critical.

Estimated Emergency Response Times

- **Urban Areas:** (Salem, Erode, Tiruppur) within 30-45 minutes.
- **Rural Sections:** Can exceed one hour depending on proximity to emergency services.

Overall Risk Assessment

- **High-Risk:** Monsoon season due to floods, specific congested areas.
- **Moderate-Risk:** Rest of the year with the need for proactive monitoring.
- **Recommendations:** Utilize GPS for real-time traffic updates, adhere strictly to road safety guidelines, and maintain frequent communication with dispatch.

In summary, while the route is manageable for transporting hazardous materials, awareness of weather patterns, traffic conditions, and infrastructure challenges is essential to minimize risk. Continuous monitoring and preparedness for emergency situations are critical to ensure the safety and efficiency of transport operations.

Risk Assessment - Turns

	Risk Type	Risk Level	Coordinates	Speed Limit
0	Turn	High	11.43968, 77.87345	15 KM/Hr
1	Turn	High	11.44029, 77.87544	15 KM/Hr
2	Turn	High	11.44898, 77.87410	15 KM/Hr
3	Turn	Medium	11.45357, 77.85789	30 KM/Hr
4	Turn	High	11.45352, 77.85698	15 KM/Hr
5	Turn	Medium	11.45847, 77.85140	30 KM/Hr
6	Turn	Medium	11.45898, 77.85135	30 KM/Hr

	Risk Type	Risk Level	Coordinates	Speed Limit
7	Turn	Medium	11.46303, 77.84945	30 KM/Hr
8	Turn	Medium	11.46314, 77.84928	30 KM/Hr
9	Turn	Medium	11.45509, 77.81383	30 KM/Hr
10	Turn	High	11.20728, 77.35914	15 KM/Hr
11	Turn	Medium	11.20246, 77.35895	30 KM/Hr
12	Turn	Medium	11.13132, 77.34190	30 KM/Hr
13	Turn	High	11.13122, 77.34181	15 KM/Hr
14	Turn	High	11.13131, 77.34033	15 KM/Hr
15	Blind Spot	Blind Spot	11.12933, 77.34007	10 KM/Hr
16	Turn	High	11.12918, 77.33333	15 KM/Hr
17	Turn	High	11.12201, 77.33196	15 KM/Hr
18	Turn	High	11.12187, 77.33169	15 KM/Hr

Emergency Locations

	type	name	coordinates	speed_limit	risk_level
3	hospital	Government Hospital	11.4525596, 77.7749426	30 km/h	Medium
4	hospital	Dhanvantri Multi Speciality Hospital	11.451362, 77.766602	30 km/h	Medium
5	hospital	Dhanvanthri Hospital	11.4496712, 77.7593772	30 km/h	Medium
6	hospital	J.K.K. Trust Hospital	11.4445841, 77.7307962	30 km/h	Medium
8	hospital	Shri Sathyanarayana Hospital	11.4291297, 77.6913408	30 km/h	Medium
9	hospital	Thanish Siddha Hospital	11.430003, 77.674964	30 km/h	Medium
10	clinic	Harshitha Clinic	11.4313207, 77.674718	30 km/h	Medium
11	hospital	Sri Kaalangi Siddhar Mooligai Vaithiya Nilayam	11.432369, 77.674894	30 km/h	Medium
12	clinic	G.K Clinic	11.4297244, 77.6749715	30 km/h	Medium
13	clinic	Erode Cancer Centre	11.3732, 77.649152	30 km/h	Medium

	type	name	coordinates	speed_limit	risk_level
15	hospital	Irt Hospital	11.2803603, 77.5644118	30 km/h	Medium
17	hospital	Gen Siddha Hospital	11.2375033, 77.5059797	30 km/h	Medium
18	hospital	Vijayamangalm Government Hospital	11.2382827, 77.501687	30 km/h	Medium
19	hospital	Dr. N Viswanathan Hospital	11.2412783, 77.5005502	30 km/h	Medium
20	clinic	P.M. Clinic	11.2281078, 77.4647296	30 km/h	Medium
22	hospital	Sri Renu Hospital	11.1990225, 77.4219937	30 km/h	Medium
23	hospital	Akshitha Hospital- Perumanallur	11.2050424, 77.3584876	30 km/h	Medium
24	hospital	Malarpriya Medical Centre - Tiruppur	11.16584, 77.351437	30 km/h	Medium
25	hospital	Jacob Hospital	11.1645096, 77.3496992	30 km/h	Medium
27	hospital	Revathi Charity Trust Hospital	11.1588612, 77.34966	30 km/h	Medium
28	hospital	Sree Saran Medical Centre	11.1488, 77.346725	30 km/h	Medium
29	hospital	Sri Kumaran Hospital, Tiruppur	11.1307528, 77.3412468	30 km/h	Medium
30	hospital	Saayi Prema, Tiruppur	11.1286682, 77.3404616	30 km/h	Medium
31	hospital	Vasan Eye Care Hospital, Tirupur	11.127834, 77.3415747	30 km/h	Medium
32	hospital	Revathy Hospital, Tiruppur	11.1202413, 77.3291542	30 km/h	Medium

Crowded Spots

	type	name	coordinates	speed_limit	risk_level
0	school	KRP Matric. Hr. Sec School	11.4546193, 77.8142445	30 km/h	Medium
1	college	Vivekanandha Engineering College	11.4589312, 77.7899284	30 km/h	Medium

	type	name	coordinates	speed_limit	risk_level
2	marketplace	Monday market	11.452863, 77.775989	30 km/h	Medium
7	school	SSM Matriculation Higher Secondary School	11.4321653, 77.6880046	30 km/h	Medium
14	college	Government polytechnic college	11.2907053, 77.5698726	30 km/h	Medium
16	school	Bharathi Matriculation School	11.2494613, 77.5307028	30 km/h	Medium
21	marketplace	Weekly Market (sandhai)	11.2287229, 77.4646676	30 km/h	Medium
26	school	Sharadha Vidhyalaya	11.1690521, 77.3485671	30 km/h	Medium

Route Photos of Risky Spots



Risk Type: Turn

Risk Level: High

Speed Limit: 15 KM/Hr

Coordinates: 11.44029, 77.87544



Risk Type: Turn
Risk Level: High
Speed Limit: 15 KM/Hr
Coordinates: 11.44898, 77.87410



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.45357, 77.85789



Risk Type: Turn
Risk Level: High
Speed Limit: 15 KM/Hr
Coordinates: 11.45352, 77.85698



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.45847, 77.85140



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.45898, 77.85135



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.46303, 77.84945



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.46314, 77.84928



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.45509, 77.81383



Risk Type: Turn
Risk Level: High
Speed Limit: 15 KM/Hr
Coordinates: 11.20728, 77.35914



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.20246, 77.35895



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.13132, 77.34190



Risk Type: Turn
Risk Level: High
Speed Limit: 15 KM/Hr
Coordinates: 11.13122, 77.34181



Risk Type: Turn

Risk Level: High

Speed Limit: 15 KM/Hr

Coordinates: 11.13131, 77.34033



Risk Type: Blind Spot

Risk Level: Blind Spot

Speed Limit: 10 KM/Hr

Coordinates: 11.12933, 77.34007



Risk Type: Turn

Risk Level: High

Speed Limit: 15 KM/Hr

Coordinates: 11.12918, 77.33333



Risk Type: Turn

Risk Level: High

Speed Limit: 15 KM/Hr

Coordinates: 11.12201, 77.33196



Risk Type: Turn

Risk Level: High

Speed Limit: 15 KM/Hr

Coordinates: 11.12187, 77.33169

